

Tests are **VERY** important.

Reinhart and Rogoff (2010)

Five countries dropped from one average.

−0.1% published. +2.2% corrected.

One row count catches it.

The agent games the check

Hardcodes the answer: 33% of the time.

Fakes a solution: 22%.

(EvilGenie, ambiguous specs. Yours are ambiguous.)

If you're not reading code, write the check.

Four checks

Check	When	Example
Data assertion	Every merge	Rows in = rows out. <code>isid id year</code> .
Known answer	Closed form	SVG to lat/lon: round-trip the four corners.
Parameter recovery	No answer	BLP: simulate from β , get β back.
Coverage	Randomness	Bootstrap: 95% interval covers truth 95% of the time.

None apply? Test what cannot change. Shuffle rows, coefficients hold. Weights of one, unweighted run.

What you type

```
# tests/test_clean.py      $ pytest
def test_merge_keeps_every_parcel():
    out = merge_owners(parcels, owners)
    assert len(out) == len(parcels)

def test_boot_se_matches_ols():
    rng = np.random.default_rng(42)
    np.testing.assert_allclose(
        boot_se(X, y, rng), ols_se(X, y),
        rtol=0.05)
```

test_*.py, test_*, assert, run pytest.
That is the framework.

Stata: assert, isid.

R: testthat.

Seed anything random.

Never == on floats.

1,000 rows, not the full run.

Freeze Table 3 and diff it.

What you tell the agent

Green checkmark

```
Write the BLP estimator  
and add some tests.
```

Checked estimator

```
Write the test first, then stop.  
Simulate 500 markets from  
beta = [-2, 1, 0.5]. Assert we  
recover it within 0.05.  
Show me the test.
```

You approve the test. The agent writes the code.

Say it in the prompt: never edit a test to make it pass.

Break the code once. A test that never fails is testing nothing.

Gentzkow and Shapiro, *Code and Data for the Social Sciences*. `pyblp/tests/test_blp.py`.